

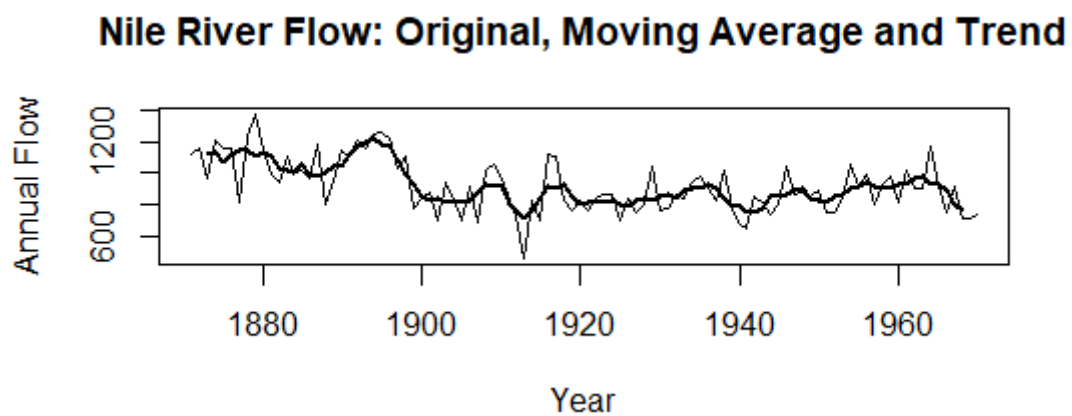
Question 1: Individual Time Series — Smoothing & Detrending

```
data(Nile)
nile_ts <- ts(
  Nile,
  start = 1871,
  frequency = 1
)
moving_avg <- stats::filter(
  nile_ts,
  filter = rep(1/5, 5),
  sides = 2
)
fitted_trend <- fitted(trend_model)
residuals <- nile_ts - fitted_trend
par(mfrow = c(2, 1))
plot(
  nile_ts,
  type = "l",
  main = "Nile River Flow: Original, Moving Average and Trend",
  xlab = "Year",
  ylab = "Annual Flow"
)
lines(
  moving_avg,
  lwd = 2
)
lines(
  time_values,
  fitted_trend,
  lwd = 2,
```

```

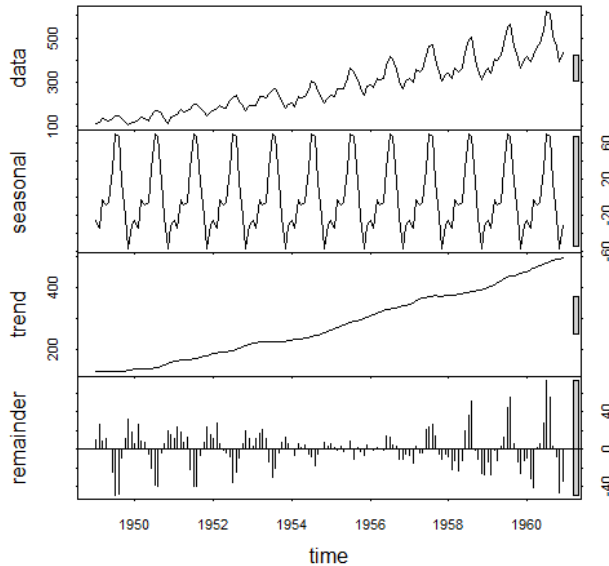
lty = 2
)
legend(
  "topright",
  legend = c("Original", "5-Year Moving Average", "Linear Trend"),
  lty = c(1, 1, 2),
  lwd = c(1, 2, 2)
)
plot(
  time_values,
  residuals,
  type = "l",
  main = "Detrended Residuals",
  xlab = "Year",
  ylab = "Residuals"
)
abline(
  h = 0,
  lty = 2
)
par(mfrow = c(1, 1))

```



Question 2: Seasonal Decomposition (STL) & Forecasting

```
library(forecast)
library(ggplot2)
data(AirPassengers)
passengers_ts <- ts(
  AirPassengers,
  start = c(1949, 1),
  frequency = 12
)
stl_result <- stl(
  passengers_ts,
  s.window = "periodic"
)
plot(stl_result)
```



Question 3: Choropleth Map & Interactive Geospatial Plot

Part a — Choropleth Map

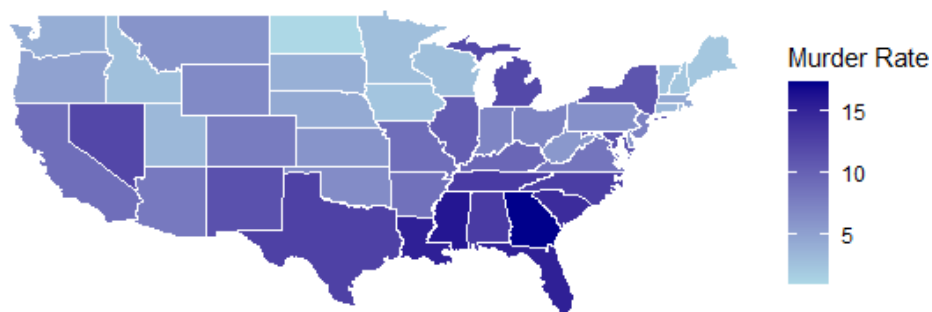
```
install.packages("ggplot2")
install.packages("maps")
install.packages("dplyr")
```

```

library(ggplot2)
library(maps)
library(dplyr)
data(USArrests)
arrests <- USArrests
arrests$State <- rownames(arrests)
rownames(arrests) <- NULL
arrests$State <- tolower(arrests$State)
states <- map_data("state")
map_data_final <- states %>%
  left_join(
    arrests,
    by = c("region" = "State")
  )
  low = "lightblue",
  high = "darkblue",
  name = "Murder Rate"
) +
labs(
  title = "Murder Rate by U.S. State",
  caption = "Source: USArrests dataset"
) +
theme_void()

```

Murder Rate by U.S. State



Source: USArrests dataset

Part b — Interactive Geospatial Plot

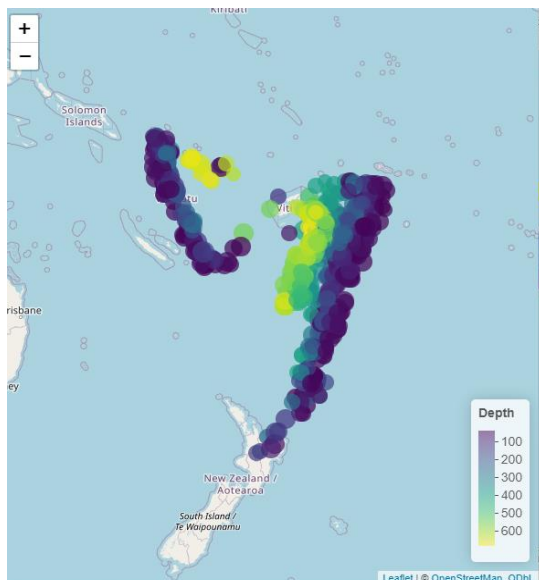
```
install.packages("leaflet")

library(leaflet)

data(quakes)

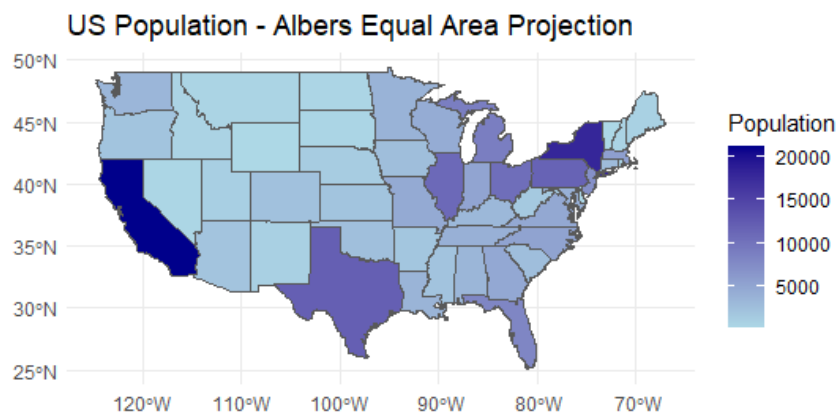
depth_colors <- colorNumeric(
  palette = "viridis",
  domain = quakes$depth
)

leaflet(quakes) %>%
  addTiles() %>%
  addCircleMarkers(
    lng = ~long,
    lat = ~lat,
    stroke = FALSE,
    popup = ~paste(
      "<b>Magnitude:</b>", mag,
      "<br><b>Depth:</b>", depth
    )
  ) %>%
  addLegend(
    position = "bottomright",
    title = "Depth"
  )
```



Question 4: Cartogram with Map Projections

```
ggplot(states_sf) +  
  geom_sf(  
    aes(fill = Population),  
    crs = 5070  
  ) +  
  scale_fill_gradient(  
    low = "lightblue",  
    high = "darkblue",  
    name = "Population"  
  ) +  
  labs(  
    title = "US Population - Albers Equal Area Projection"  
  ) +  
  theme_minimal()
```



Question 5: 3D Geospatial Heatmap

```
install.packages("plotly")
install.packages("ggplot2")
library(plotly)
volcano <- read.csv("volcano_elevation.csv")
elevation_grid <- xtabs(elevation ~ y + x, data = volcano)
fig <- plot_ly(
```

```

x = colnames(elevation_grid),
y = rownames(elevation_grid),
z = elevation_grid,
type = "surface",
colorscale = "Earth"
)
fig <- fig %>%
  layout(
    title = "3D Geospatial Heatmap of Maunga Whau Volcano",
    scene = list(
      xaxis = list(title = "X"),
      yaxis = list(title = "Y"),
      zaxis = list(title = "Elevation")
    )
  )
)

```

